

MRITUNJAY SHARMA | AI & Machine Learning Engineer

mritunjaysharma080505@gmail.com +91 9805355446 [LinkedIn](#) [GitHub](#)

Summary

AI & Machine Learning Engineer with project-driven experience in Deep Learning, Generative AI, and serverless cloud-based ML systems. Skilled in designing LLM-powered applications, Hybrid Agentic RAG pipelines, and AI pipeline deployment. Strong foundation in Probability, Statistics, and Neural Network Optimization, with hands-on expertise in AWS, PyTorch, FAISS, and end-to-end AI solutions. Published researcher in Retrieval-Augmented Generation systems.

Education

- **Associate of Science in Artificial Intelligence | Saras AI Institute, USA** (Expected 2026).
 - Relevant Coursework: AI Fundamentals, Python with Version Control, Probability & Statistics, Math for AI, Cloud Fundamentals, AI in Cloud, Data Analysis with Python, Machine Learning.
- **Bachelor of Computer Applications (BCA) | MCM DAV College, Himachal Pradesh University** .

Publications

- **Moksh AI v3.0: An Agentic Hybrid Retrieval-Augmented Generation System with Explicit Query Routing and Dual-LLM Architecture**
 - Mritunjay Sharma. Preprint, Zenodo, 2026.
 - Proposed explicit agentic retrieval decision layer achieving 92% accuracy vs 64% baseline (+28pp) on 25-question benchmark across 5 categories.
 - Designed dual-LLM architecture (Llama 3 local + Gemini 2.0 Flash cloud fallback) with autonomous background memory extraction.
 - Link: zenodo.org/records/20444127

Projects

Moksh AI v3.0 — Hybrid Agentic RAG System

- Built locally-running AI assistant with agentic decision layer routing queries between FAISS vector store, web search, or hybrid mode using temporal keywords and L2 distance thresholds (CONF_HIGH=0.50, CONF_LOW=1.50).
- Implemented dual-LLM resilience with Ollama Llama 3 (local) and automatic Gemini 2.0 Flash cloud fallback ensuring zero-downtime operation regardless of hardware availability.
- Developed Streamlit web app with voice input (Whisper STT), vision support (Llama 3.2-vision), multi-session management, auto memory extraction, and chat export. Published research paper — 92.0% accuracy on 25-question benchmark.

Deepfake Face Detector — Real-Time Computer Vision Forensic System

- Developed binary classification model using PyTorch and ResNet-50 with custom classification head (2048→512, Dropout 0.3) achieving 94.05% test accuracy and ROC-AUC ~0.98.
- Trained on 140,000 high-resolution images using Mixed Precision Training (torch.cuda.amp) on NVIDIA RTX 4060 completing 5 epochs in ~30 minutes.
- Evaluated model rigorously beyond accuracy — Precision 0.95, Recall 0.93 ensuring minimal false alarms on real faces and maximum forgery detection.
- Built Streamlit interface for real-time image and video forgery detection with <12ms inference latency per frame.

Customer Churn Prediction API

- Built end-to-end ML system on IBM Telco dataset (7,032 customers) with full sklearn Pipeline (SimpleImputer + OneHotEncoder + StandardScaler + Logistic Regression) achieving ROC-AUC 0.836 and 80% accuracy.
- Deployed as production REST API with Flask — POST /predict endpoint returning churn probability and binary prediction with <10ms average response latency.
- Implemented overnight batch scoring pipeline processing 100+ customers with 0 failures and average churn probability of 27.2%.
- Designed monthly retraining strategy with data drift detection and date-stamped model versioning for production maintenance.

FashionNet — Deep Neural Network Optimization

- Built and optimized a 3-block Convolutional Neural Network for fashion item classification using PyTorch, implementing BatchNorm, Global Average Pooling, and Cosine LR scheduling.
- Improved test accuracy from 88.2% → 93.15% through systematic architectural tuning, dropout regularization (0.3), label smoothing, and gradient clipping over 20 epochs.
- Conducted detailed error analysis identifying Shirt as highest confusion class (95 misclassifications vs T-shirt/top) — inherent visual similarity confirmed via confusion matrix.
- Executed controlled experiments with best model checkpointing, JSON training logs, and full evaluation suite — Precision 0.93, Recall 0.93, F1 0.93 across 10 classes.

Skills

- **Core Competencies:** Machine Learning, Deep Learning, Generative AI, RAG Systems, Agentic AI, Prompt Engineering, Neural Network Optimization, AI Pipeline Deployment, Serverless AI Workflows, Data Analysis
- **Frameworks & Libraries:** PyTorch, Scikit-learn, TensorFlow, FAISS, Sentence Transformers, NumPy, Pandas, Streamlit, Ollama, Llama 3, Gemini 2.0 Flash, VADER, NLP (Tokenization, Embeddings, Lemmatization)
- **Cloud & Deployment:** AWS Lambda, API Gateway, CloudWatch, IAM, Docker, Flask, FastAPI, Hugging Face Spaces
- **Tools & Platforms:** Google Colab, Git, GitHub, Jupyter Notebook, DuckDuckGo Search API